

An Open Firm-Level Microsimulation for UK VAT Threshold Policy Analysis*

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Abstract

I present an open firm-level microsimulation for transparent, conditional costing of UK VAT registration-threshold reforms. The model generates a weighted accounting population from public aggregates; it does not recover administrative microdata or firm behaviour. On a common 2023–24 baseline, I apply *level*, *shape* (a taper), and *rate* (a reduced-rate band) scenarios and report their mechanical revenue effects. I characterise the notch by its model-implied *dominated region*—a £21,250 band above the £85,000 threshold—and show how each schedule changes it. Intensive-margin results are sensitivity exercises conditional on a region-confined response model and assumed elasticities; the extensive margin is not identified. The near-threshold shape is target-inherited, and specification diagnostics show that the synthetic data cannot support bunching inference. The exercise therefore supplies reproducible policy scenarios and a general warning: calibrated synthetic data cannot create behavioural or sub-band fiscal information absent from their targets and maintained assumptions.

Keywords: value-added tax, microsimulation, tax notch, registration threshold

JEL codes: H25, H26, C63

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1 Introduction

The United Kingdom’s Value Added Tax (VAT) applies above a registration threshold that is the highest in the OECD alongside Switzerland (Seely, 2024; HM Revenue and Customs, 2024). The threshold is a *notch* rather than a kink: once a firm’s annual taxable turnover crosses it, VAT falls due on the firm’s whole base, not only on the part above the threshold, so registration produces a discrete jump in liability rather than a marginal one. The threshold falls in a dense part of the firm distribution. Figure 1(a) shows that VAT- and PAYE-registered UK businesses are concentrated at the lower end of the turnover distribution, and Figure 1(b) shows that the VAT-registered population per £1,000 of turnover is densest at and below the threshold, so the notch applies to a large number of firms.

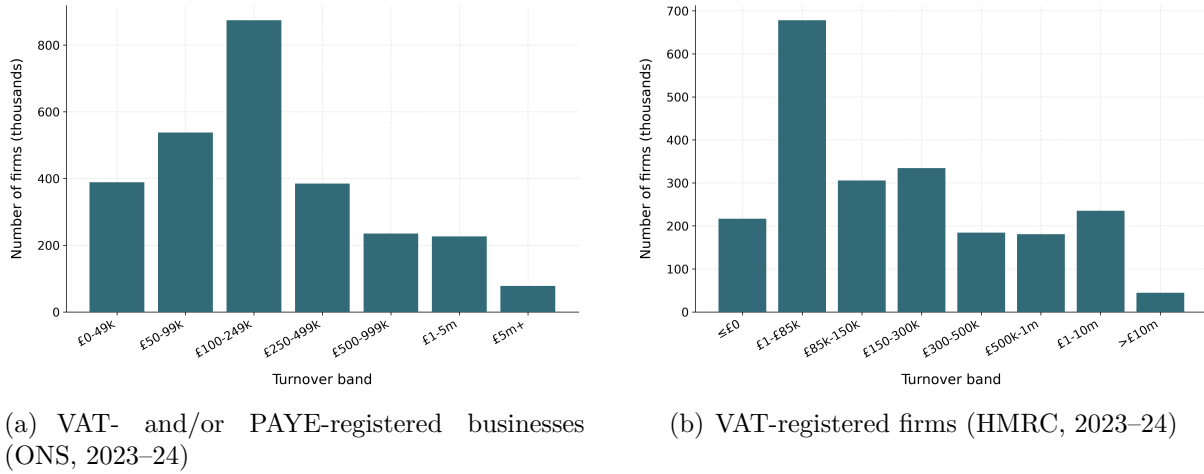


Figure 1: The UK firm-size distribution and VAT registration, 2023–24 (threshold £85,000). Panel (a): number of businesses by turnover band in the ONS business register, which covers businesses registered for VAT and/or PAYE (about 2.7 million); the full private-sector business population including the 2.9 million businesses registered for neither is about 5.5 million (DBT business population estimates, start of 2024). Panel (b): number of VAT-registered firms by turnover band. Band widths are unequal, so cross-band count comparisons should be read per £1,000 of turnover. Source: panel (a), ONS, *UK Business: Activity, Size and Location*, 2024 edition (Table 8, enterprises by turnover sizeband); panel (b), HMRC, *Value Added Tax (VAT) Annual Statistics* (VAT-registered population by turnover band).

Firms respond to the notch by locating below the threshold. Tabulations of HMRC turnover data show the number of firms rising towards £84,000 and then dropping sharply at the £85,000 threshold, leaving excess mass just below it (Office for Budget Responsibility, 2023). Liu et al. (2021) document this bunching in UK administrative firm data over 2004–05 to 2014–15, when the threshold rose from £58,000 to £81,000, and Liu et al. (2024) find that firms slow their turnover growth as they approach the threshold. The

notch therefore distorts the size distribution and the growth decisions of firms near the margin, which places the threshold under recurrent pressure for reform.

Reform proposals change the threshold in one of three ways: its *level* (raising or lowering it), its *shape* (replacing the notch with a schedule that phases the rate in over a band), or its *rate* (a reduced rate for firms in a band above the threshold). The three are not equivalent. Changing the level moves the notch along the turnover axis; changing the shape or the rate alters the size of the discrete jump itself. Costing the three on a common basis is the practical difficulty this paper addresses. A change in the level can be costed by mechanical accounting, but a change in the shape of the schedule cannot be evaluated with the reduced-form bunching methods used to study the threshold, because a continuous schedule leaves no single point of excess mass to exploit. The firm-level data needed to study turnover near the threshold is, in addition, not available in open form.

In this paper I build an open firm-level accounting microsimulation for conditional costing of UK VAT threshold reforms on a common static basis. I construct weighted synthetic firm records calibrated to published official aggregates—HMRC registered-firm counts by turnover band and by sector, net VAT liability by turnover band, and ONS business counts and employment bands—for the 2023–24 tax year, when the threshold was £85,000, and separately for 2024–25, when it rose to £90,000. These are not reconstructed administrative records: turnover and liability within published bands are simulated under explicit assumptions. The released code permits those assumptions and counterfactual schedules to be changed and audited.

On these data I do four things. First, I cost level reforms and schedule reforms—a graduated taper and a reduced-rate band—statically, by applying each counterfactual schedule mechanically to the firm distribution. Second, I characterise the notch’s distortion through its *dominated region*—the exact, elasticity-free range of turnover above the threshold in which no firm profits from locating—and how each reform changes it. Third, I price the level and rate reforms behaviourally: each firm re-optimises its turnover within its schedule region, conditional on an assumed turnover elasticity e that the data do not identify, reported as an e -sensitivity range that nests the static costing exactly in its $e \rightarrow 0$ limit. Fourth, I show that the synthetic data’s near-threshold structure is target-inherited and cannot support bunching inference: the £85,000 profile is calibrated to the OBR’s published £1,000-band data and the generator contains no location-choice mechanism.

Four findings follow. First, the static pipeline prices level, shape, and rate reforms on one basis; its anchor—the April 2024 rise from £85,000 to £90,000, whose mechanical cost profile lands within 30% of HMRC’s published costing in every release year and within 10% in 2025–26 and 2026–27, with no retention or behavioural parameter tuned—is a consistency check rather than an out-of-sample validation, since the population is calibrated to HMRC aggregates (Section 4). Second, the shape and rate reforms act on the notch’s distortion in a way a level change does not: the notch creates an exact £21,250 dominated region, which a rise relocates and widens, a reduced-rate band splits across two notches, and only a continuous taper removes—and the taper’s band width is a property of its shape, not a requirement of removal (Section 6). Third, conditional on the assumed elasticity, intensive-margin responses barely move the costings (Section 7).

Fourth, the synthetic data carry a transparency lesson: their near-threshold excess mass is inherited from calibration targets or band edges, is unstable across specifications, and cannot validate a bunching estimator or identify behaviour (Section 5); the behavioural evidence remains the administrative-data work of [Liu et al. \(2021\)](#). Table 4 collects the headline estimates.

Related literature

The paper draws on four strands of work. The first is the methodology of bunching and notches ([Chetty et al., 2011](#); [Saez, 2010](#); [Kleven and Waseem, 2013](#); [Kleven, 2016](#); [Best et al., 2015](#)), which provides the analytic notion of a dominated region; I use the dominated region to characterise a distortion that leaves no single point of excess mass to exploit. Recent evidence from a Finnish payroll-tax notch shows that a policy threshold can impede mobility and firm growth well beyond the immediate cutoff and can bias conventional difference-in-differences comparisons ([Jysmä et al., 2026](#)). That result strengthens the case for treating my local dominated-region metric as partial: its payroll and employment setting does not identify the response to the UK VAT turnover threshold, but it demonstrates the potentially wider dynamic margin that this accounting model omits.

The second is the empirical literature on VAT registration thresholds ([Liu et al., 2021, 2024](#); [Onji, 2009](#); [Harju et al., 2019](#); [Asatryan and Peichl, 2017](#); [Nandi and Warwick, 2020](#); [Bashir et al., 2024](#)). [Liu et al. \(2021\)](#) is closest: it estimates bunching at the UK notch on administrative data over 2004–2014 and documents voluntary registration; I take its estimates as the established behavioural facts and cost shape and rate reforms statically on a common basis, which that literature does not do. The international evidence ([Harju et al., 2019](#); [Onji, 2009](#); [Asatryan and Peichl, 2017](#); [Nandi and Warwick, 2020](#)) notes that bunching at a VAT threshold can reflect compliance costs, firm-splitting, or reporting rather than real turnover responses.

The third strand questions whether bunching identifies a structural elasticity. [Blomquist et al. \(2021\)](#) show that it is not non-parametrically identified at a kink or notch without strong functional-form assumptions; [Bertanha et al. \(2023\)](#) develop the notch case. I therefore do not infer an elasticity from the synthetic data. The fourth strand is the theory of size-based tax thresholds: the optimal VAT threshold of [Keen and Mintz \(2004\)](#), thresholds as partitions of compliance in [Kanbur and Keen \(2014\)](#), and optimal remittance with firm-level administrative costs in [Dharmapala et al. \(2011\)](#), together with the structural literature on size-dependent policy ([Garicano et al., 2016](#); [Gourio and Roys, 2014](#)); the Mirrlees Review’s VAT chapter frames the UK debate ([Crawford et al., 2010](#)). The present paper makes no comparable structural claim. Firm-level tax microsimulation has precedents in corporate-tax models such as the German BizTax model ([Fossen and Bach, 2008](#)); to my knowledge no open firm-level model of the UK VAT threshold exists, which is the gap this paper fills.

This paper makes three contributions. First, it provides an open and auditable firm-level accounting microsimulation for conditional VAT-threshold scenarios, where the existing behavioural evidence rests on restricted administrative data. Second, it characterises the notch through its model-implied dominated region and costs two schedule reforms—

a graduated taper and a reduced-rate band—on the same static basis as level reforms. Third, it demonstrates that band-calibrated synthetic data carry no sub-band information beyond their targets and distributional assumptions. All fiscal estimates below should therefore be read as conditional scenario outputs, not point forecasts with administrative-data precision. Identifying the turnover elasticity from administrative microdata, welfare accounting, revenue-neutral designs, and an explicit model of voluntary registration are left for future work.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background of the UK VAT threshold. Section 3 sets out the synthetic firm microdata and its calibration to official aggregates. Section 4 sets out the static costing of threshold changes and the schedule reforms. Section 5 shows that the synthetic data contain no usable behavioural bunching signal. Section 6 develops the notch and its exact dominated region, and shows how each reform changes it. Section 7 prices the reforms behaviourally, conditional on an assumed elasticity, and derives the level-rise invariance. Section 8 concludes.

2 Institutional background

Registration and the notch. The United Kingdom levies a value-added tax at a standard rate of 20% on most goods and services. A business must register once its taxable turnover over any rolling twelve-month period exceeds the registration threshold, after which it charges VAT on its taxable sales and reclaims VAT paid on inputs. Because registration makes VAT due on the firm’s *entire* taxable turnover rather than only the amount above the cutoff, the threshold is a *notch*—a discrete jump in liability—and not a kink. The cost of crossing is largest for consumer-facing firms with limited pass-through and small reclaimable input shares, for whom the incentive is to restrain turnover and locate just below the threshold (Liu et al., 2021; Kleven and Waseem, 2013).

Value added, input reclaim, and voluntary registration. Real VAT is a tax on *value added*, not on turnover: a registered firm charges output VAT on its sales but reclaims input VAT on its purchases. The model reflects this: each synthetic firm’s net liability is the standard rate applied to its value added (Section 3), and under the value-added formulation—in which unregistered firms bear no unreclaimed input VAT—the notch’s dominated region is invariant to the firm’s input share (Section 6). Two features of the real system remain outside the model’s scope. A large share of below-threshold firms register *voluntarily*—roughly 43% of eligible below-threshold firms (Liu et al., 2021), driven by input-VAT reclaim and business-to-business sales—and for them the threshold carries no bunching incentive; voluntary registration enters the model only as a registration-rate parameter and a retention sensitivity, not as a choice. And net-repayment traders (zero-rated exporters and similar, HMRC’s negative-liability column) are not represented, because the public aggregates do not resolve rated structure by turnover band.

Threshold history. The registration threshold was held at £85,000 from 1 April 2017 to 31 March 2024—a seven-year nominal freeze, the longest in the history of the tax—before being raised to £90,000 on 1 April 2024, with the deregistration threshold rising from £83,000 to £88,000 (HM Revenue and Customs, 2024; Seely, 2024). Because nominal turnover grows with inflation and real activity, freezing the threshold lowers it in real terms and draws an increasing number of firms towards the registration margin; this fiscal-drag mechanism was a central motivation for the April 2024 increase. My data year is 2023–24, so the prevailing statutory threshold throughout my analysis is £85,000. Figure 2 shows the pattern in the turnover distribution: the number of firms rises towards £84,000 and falls sharply above £85,000, and the projected 2025–26 distribution is more sharply bunched than the 2019–20 outturn as fiscal drag draws more firms towards the frozen threshold.

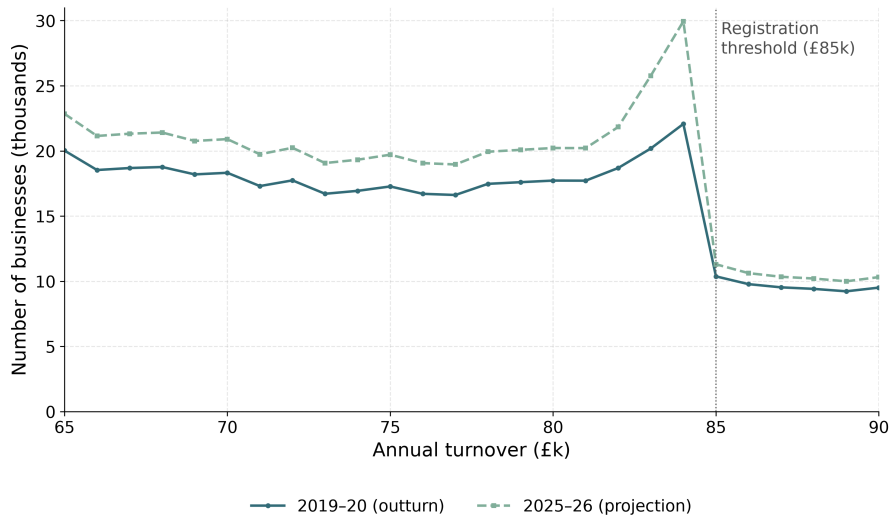


Figure 2: Bunching below the £85,000 VAT registration threshold: number of businesses by £1,000 turnover band. Source: OBR, *Economic and Fiscal Outlook March 2023*, Chart C (Open Government Licence).

The reform debate. The UK threshold is the highest in the OECD alongside Switzerland (HM Revenue and Customs, 2024), and there is debate over whether to raise it—to reduce the compliance burden on small firms—or lower it, to broaden the base and reduce the growth distortion at the margin. Three families of reform feature, and I evaluate all three: a change in the threshold’s *level* (the £85,000 to £90,000 increase of April 2024 and further moves around it); a change in its *shape* (a *graduated taper* that phases the effective rate in over a band, replacing the notch with a continuous profile); and a change in its *rate* (a *reduced rate* for small firms in a band above the threshold, with the standard 20% rate retained above). The taper and reduced-rate designs follow proposals in the UK policy debate: Office of Tax Simplification (2017) canvassed a financial incentive or reduced effective rate to smooth the cliff-edge at the threshold, and HM Treasury’s call for evidence on the registration threshold examined smoothing mechanisms in the

same family (HM Treasury, 2018). The European Union ran a closely related “graduated relief” scheme for small enterprises and abolished it from January 2025 on complexity grounds, in favour of a simple exemption with a transition tolerance (Council of the European Union, 2020)—relevant precedent for the administrability of taper-style designs. A banded reduced rate introduces a smaller *secondary* notch at the band top, which the continuous taper avoids. I cost all three by the same machinery in Section 4.

3 Data

Studying firm responses at the threshold needs firm-level turnover near the cutoff, which the official statistics publish only in coarse turnover bands. I therefore build a synthetic firm population calibrated to those aggregates: it reproduces selected official totals while supplying explicitly simulated within-band records for accounting exercises. These records are not estimates of the confidential firms that the authorities do not release. The same calibrate-to-published-aggregates approach underlies open household-level tax-benefit microsimulation (PolicyEngine, 2025); here I apply it at the firm level. The generator takes the registration threshold as a parameter and is documented in Appendix A.1; I build one population for 2023–24 (threshold £85,000), the paper’s primary data year, and one for 2024–25 (threshold £90,000), used for the forward sweep of Section 4.

The archived CSV inputs in this repository remain the reproduction source for the paper’s reported numbers; a pinned migration snapshot also represents the same numeric target surface through PolicyEngine Ledger and the experimental Populace firm generator, and Appendix A.1 reports the parity check between that shared source-of-truth path and the paper inputs.

Construction. Index firms by i . For each sector s and ONS turnover band $b = [\underline{b}, \bar{b})$, the *UK Business: Activity, Size and Location* table gives a business count $N_{s,b}$, and I draw $N_{s,b}$ firms with turnover inside the band. The within-band density is a maintained assumption, not an observed distribution. In the bottom band ($[0, 50)$ thousand pounds) the draw is uniform. In each closed band above it the draw is a truncated power law, $f(y) \propto y^{-\alpha_b}$ on $[\underline{b}, \bar{b})$, with α_b the log–log slope of the band-average densities of the two neighbouring bands (a central difference, one-sided at the ends), so the density is monotone inside every band and approximately continuous across band edges; the exponents are 0.07, 1.21, 1.68, 1.79, 1.84 for the bands from £50,000 upward. The open “£5m and over” band is drawn uniform in log turnover on $[\$5\text{m}, \$50\text{m})$. Band counts are preserved exactly under every choice; a uniform fill, used in earlier drafts, left a visible density step at each ONS band edge, including one at £100,000 inside the region the reform costings depend on. I interpret integer-labelled bands as contiguous half-open intervals: for example, the published “50–99” band is represented as $[50, 100)$ thousand pounds. The counts come from Table 8 of the ONS workbook (enterprises by SIC division and turnover sizeband, United Kingdom); the employment-band targets below come from Table 3 of the same workbook (enterprises by employment sizeband), so both ONS margins count

the same statistical unit.¹

Three universes. The model distinguishes three populations and declares which one each calibration target describes. The *frame* is the ONS register of enterprises operating VAT and/or PAYE—about 2.72 million in 2024—which the population and employment targets describe. The *registered subset* is the population of HMRC VAT traders—about 2.18 million in 2023–24—which the HMRC counts by turnover band and by trade sector and the net-liability totals describe. The two differ in both directions. Above the threshold only about two-thirds of frame enterprises are VAT traders (from 54% in the £85,000–£150,000 band to 94% in the £1m–£10m band and 82% above £10m): the remainder are PAYE-only employers and businesses in exempt sectors (finance, health, education, and similar) outside the VAT net. The two top bands depend on how the open ONS “£5m and over” band is filled, since it has no published top; I draw it uniform in log turnover on [\$5m, \$50m), the maximum-entropy choice for an open band, and the band’s liability total is calibrated regardless, so reform costings near the threshold are unaffected. Below the threshold about 89% of frame enterprises are registered, largely because small businesses enter the register *through* VAT registration. HMRC also reports about 216,500 traders with negative or zero turnover who are not identifiable in the ONS turnover bands; I append them as an out-of-frame stratum before calibration. The third universe is the *unregistered stratum*: businesses registered for neither VAT nor PAYE, which the DBT Business Population Estimates count at about 2.86 million at the start of 2024, with total turnover of £123bn (a mean of £43,000) and, by SIC division, both the count and the turnover total (Department for Business and Trade, 2024). Together the three universes make up the DBT private-sector business population of about 5.5 million. BPE publishes no size distribution for the stratum, so each division’s turnover is drawn from an exponential distribution with the division’s mean—the maximum-entropy density for a positive variable with a known mean—and the stratum’s weights are held fixed in calibration: its shape is a maintained assumption, not a fitted object. Businesses in the stratum below the threshold are registrable (they would have to register on crossing it); those the draw places above it can only be exempt-sector traders and are placed outside VAT scope. The stratum changes what the OBR near-threshold data can pin down. Chart C counts every business in HMRC records by £1,000 band, a universe that below the threshold is the frame plus the unregistered stratum; with the stratum present I apply the chart’s *levels* as targets—on frame plus stratum below the threshold and on the frame alone above it, where unregistered businesses are exempt traders outside the chart’s registration analysis—so that the frame’s near-threshold density is identified as the residual between the chart and the stratum, rather than the frame-scaled shape targets of earlier drafts. Each firm is given input expenditure $x_i = \rho_i y_i$, with the input–output ratio ρ_i drawn from a rescaled Beta distribution with sector-specific shifts, clamped to $\rho_i \in [0.1, 0.95]$ (mean about 0.6, so value added averages about 40% of turnover, in line

¹An earlier build of the 2023–24 population used the local-unit table (Table 18, individual sites: 3.17 million against 2.72 million enterprises) for the employment margin, so its employment and population targets could not be met by the same weights. The processed tables are now produced by an executable extract from the raw workbooks and tested for agreement to disclosure rounding.

with the UK non-financial business economy). Its net VAT liability is the standard rate applied to its value added,

$$v_i = \tau (y_i - x_i), \quad \tau = 0.20,$$

so every firm’s net rate $v_i/y_i = \tau(1 - \rho_i)$ lies in $[1\%, 18\%]$, strictly positive and below the statutory cap of 20%. The model therefore abstracts from net-repayment traders (firms with zero-rated outputs whose input reclaim exceeds output VAT—HMRC’s negative-liability column); representing them requires rated-structure detail (zero, reduced, and exempt output shares) that the public aggregates do not resolve by band. Employment is assigned from the ONS employment-band shares conditional on each firm’s sector, and the same assignment is retained through calibration and final output. Appendix A.1 records the model and data provenance relevant to replication.

The base population reproduces the ONS structure but not the HMRC VAT-registered totals, so I re-weight it. Each firm receives a strictly positive weight $w_i = e^{\theta_i}$, and for a vector of official targets $T = (T_k)_{k=1}^K$ the weighted synthetic counterpart of target k is

$$\hat{T}_k(\theta) = \sum_i a_{ki} w_i,$$

where a_{ki} is firm i ’s contribution to target k on the universe the target describes. For the frame targets—the ONS enterprise total and the employment bands— a_{ki} is a membership indicator for frame firms and zero for the appended HMRC traders. For the registered-subset targets—HMRC trader counts by turnover band and by sector, and net liability by band— a_{ki} is the firm’s *registration propensity* $p_{b(i)}$ times the band or sector indicator (times v_i for a liability target), where p_b is the HMRC trader count in turnover band b divided by the frame’s mass in that band; appended traders carry $p = 1$. At unit weights the registered-subset rows therefore reproduce the HMRC counts by construction and the frame rows reproduce the ONS counts by construction; the optimiser’s work is to fit the sector composition, the liability totals, and the near-threshold shape while holding both. I choose θ to minimise a symmetric relative-error loss with per-target importance weights λ_k ,

$$L(\theta) = \frac{1}{K} \sum_k \lambda_k \min\left\{(\hat{T}_k/T_k - 1)^2, (T_k/\hat{T}_k - 1)^2\right\} + \frac{\alpha}{N} \sum_i |\theta_i|,$$

the final term a mean absolute penalty on the log-weights. The targets are the HMRC VAT-registered counts by turnover band (all eight bands, including the negative-or-zero band) and by trade sector, the HMRC net VAT liability by turnover band, the ONS enterprise total and employment-band totals, and—for the 2023–24 vintage—the OBR’s published near-threshold profile. HMRC’s sector table includes traders whose turnover band is “Unknown” (159,300 in 2024–25, none in 2023–24); I allocate them proportionally so the sector and band rows count the same registered population. Chart C of the March 2023 *Economic and Fiscal Outlook* reports HMRC business counts in £1,000 turnover bands from £65,000 to £90,000 ([Office for Budget Responsibility, 2023](#)). I interpolate these counts to 2023–24 between the 2019–20 outturn and the OBR’s 2025–26 frozen-threshold projection. The final £90,000 row lies beyond the fine window and is not used.

All 25 retained bins enter as *shape* targets. I apply them separately on each side of the threshold: the below-threshold profile is normalised over £65,000–£85,000 and the above-threshold profile over £85,000–£90,000. Each is then scaled to the synthetic population’s mass in its respective window. Chart C counts HMRC traders, a different unit and, below the threshold, a substantially narrower population than the ONS business frame, so its levels are not imported on either side. An earlier build that mixed conventions (direct counts above the threshold, frame-scaled shape below) placed more mass just above the threshold than just below it, inverting the administrative profile’s cross-threshold ordering; under side-consistent scaling the cross-threshold step comes from the frame’s own band structure and the OBR data supply only the within-side geometry—the rise into the threshold and the decline beyond it. These fine targets pin the near-threshold density to the OBR profile’s within-side geometry at frame-implied levels—the region that coarse bands alone leave to the weight optimiser (Section 5); the resulting cross-threshold step (about -21% between the two bins adjoining the threshold) is about a third of the OBR chart’s own (-60%), because the chart’s below-threshold universe is narrower than the frame. No fine bands are published for the £90,000 threshold era, so the 2024–25 vintage keeps coarse bands only. Turnover-band and near-threshold targets carry the largest λ_k , followed by VAT liability by band. Two liability dimensions are reported as informational diagnostics but excluded from the optimizer. VAT liability by sector is excluded because the generator draws a common input-share distribution with sector shifts rather than calibrating the input/output tax structure by sector. The below-threshold liability band (£1–threshold, £1.46bn in 2023–24) is excluded because that total is remitted by voluntary registrants—input-reclaim traders averaging about £2,150 of net remittance—whom the model’s standard-rate-on-value-added liability does not represent (Section 2); imposing it on the whole below-threshold population would force the optimiser to suppress weights in the £50,000–£85,000 region, distorting the near-threshold shape the fine targets exist to pin. The problem is solved by gradient descent, and the symmetric loss keeps targets of very different magnitudes on a common scale.

Scope and registration. After calibration each firm receives two flags. *Scope* marks firms in the VAT net: above the threshold, a seeded weighted selection within each HMRC turnover band whose cumulative weight reaches that band’s HMRC trader count (the remainder are out-of-scope PAYE-only or exempt-sector enterprises, which remit nothing under any schedule considered here); below the threshold, every frame enterprise is treated as registrable, a maintained assumption since HMRC publishes only the registered count there. *Registration* marks in-scope firms above the threshold and, below it, a seeded weighted selection reaching the HMRC £1–threshold count; appended negative-or-zero traders are in scope and registered; unregistered-stratum businesses are in scope below the threshold and never registered at the data year. Registration above the threshold is thus certain for in-scope firms. This is an approximation to the statutory test, which is a rolling 12-month test on *taxable* turnover with a roughly two-month registration lag, a forward-looking 30-day test, and an exception for temporary breaches (Section 2); exempt-sector businesses are represented only through the out-of-scope share, not identified individually.

For a single-year cross-section calibrated to registered-population totals these timing and composition wedges are second order for the aggregates, but they are worth naming: the HMRC band tables classify traders by total declared outputs (VAT-return Box 6), and the ONS turnover measure is likewise total turnover, so the running variable throughout is total rather than taxable turnover. The seeded selections match each weighted total to within one firm’s weight but do not identify *which* firms are in scope or register voluntarily; their composition remains a modelling assumption. The resulting dataset comprises approximately 5.80 million firm records per vintage (2.72 million frame draws, 2.86 million unregistered-stratum businesses, and 0.22 million appended traders), weighted to represent about 2.72 million frame enterprises in 2023–24, of which about 2.18 million are VAT-registered traders, plus the 2.86 million unregistered businesses at fixed weight. Checked against the official sources along every calibrated dimension, the vintages attain overall calibration accuracies of 98.3% (2023–24) and 98.2% (2024–25); Appendix A.1 reports the full table. The headline is the mean of five calibrated-dimension scores, each clipped at zero as $\max\{0, 1 - |\hat{T} - T|/|T|\}$, HMRC dimensions scored on registered rows and ONS dimensions on frame rows; it excludes the VAT-liability-by-sector diagnostic described above. Because the turnover-band rows hold by construction, the informative residuals are the sector composition (93.3% and 92.6%) and the liability totals by band (98.4% and 98.9%).

Aggregate fit does not imply record-level representativeness. Kish effective sample sizes over the calibrated rows (the frozen stratum excluded) are 1,091,716 (37.1%) and 1,214,074 (41.2%), while maximum weights are about 298 and 568; the median weight is one. The log–log within-band fill removed the extreme weights (above 2,000) that the uniform fill had needed in the large-turnover bands. Synthetic rows should therefore be read as weighted accounting support points, not independent firm observations; local and sub-band results can be sensitive to a small number of high-weight records. I report rounded scenario estimates and expose the full weight diagnostics in the generated calibration report.

The registration threshold in the data. My primary data year is 2023–24, when the statutory threshold was £85,000 (Figure 3 shows the calibrated distribution). The HMRC band targets step down across the threshold—678,350 firms in the £1–£85,000 band against 305,320 in £85,000–£150,000, or roughly 8,000 against 4,700 firms per £1,000 of band width—and the registered subset of the calibrated population inherits that band-average step through the registration propensity; the frame density itself steps only from about 9,000 to 8,700 enterprises per £1,000 across the same bands. The *local* shape within a few thousand pounds of the threshold is pinned by the OBR near-threshold targets described above: the calibrated all-business density rises into the threshold (about 27,300 firms per £1,000 in the £84,000–£85,000 bin, of which 18,600 are frame enterprises and 8,700 unregistered businesses), steps down across it (about 19,600 just above), and declines through the pinned window—the administratively observed bunching geometry (Office for Budget Responsibility, 2023; Liu et al., 2021), present in the file because it is targeted, not because any synthetic firm chooses it. In an earlier build without these targets the local shape was an unpinned optimiser equilibrium (and, under the

pre-correction liability scale, a spurious bunching look-alike); the central data caveat of the paper stands in all three configurations: no sub-band feature of band-calibrated synthetic data carries behavioural information beyond what its targets put there (Section 5 quantifies the resulting specification sensitivity). Counterfactual policies—including the £85,000 to £90,000 reform and the forward threshold sweep of Section 4—are evaluated on these firms by re-applying the registration rule to the microdata for the static costings, holding the empirical turnover distribution fixed. A behavioural layer that re-prices firms under counterfactual schedules, conditional on an assumed turnover elasticity, is developed in Section 7.

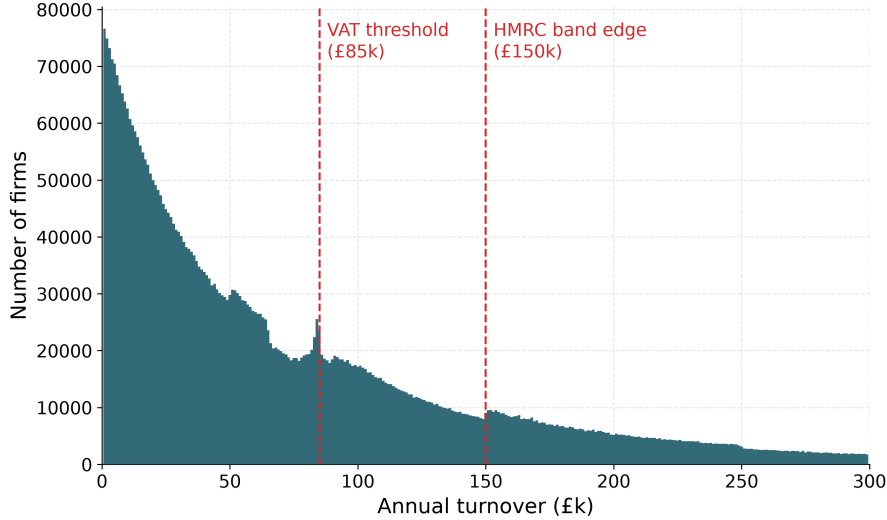


Figure 3: Synthetic firm turnover distribution (weighted firm counts, £1,000 bins), 2023–24. The near-threshold profile—rising into the £85,000 threshold and stepping down across it—is calibrated to the OBR’s published £1,000-band counts and is therefore target-inherited, not behavioural (Section 5). Levels are the all-business universe: the ONS VAT/PAYE frame plus the DBT unregistered stratum below the threshold, the universe of the OBR chart, so the cross-threshold step is comparable to the administrative one. The steps at £65,000 (the lower edge of the OBR-pinned window, below which no fine targets exist), at £90,000 (about 17,800 firms per £1,000 in the last OBR-pinned bin against about 19,300 in the first unpinned bin, where the density reverts to the coarse-band level) and £150,000 are calibration-target boundaries—the top of the OBR fine-band window and an HMRC band edge respectively—not features of the underlying firm distribution. The £90,000 boundary lies inside the bunching estimator’s fit range; Appendix A.2 measures its effect on the headline statistics.

4 Static costing of threshold reforms

A static costing measures the first-order revenue effect of a reform by re-applying the tax rule to a fixed firm population, with no behavioural response. Let firm i have turnover y_i ,

calibrated net VAT liability $v_i = \tau(y_i - x_i)$ (Section 3), weight w_i , scope flag s_i (one for firms in the VAT net), data-year registration flag q_i , and baseline voluntary-registration flag r_i (one for firms registered with data-year turnover at or below the data-year threshold T_0). A firm remits VAT only once registered. Under a counterfactual threshold T an in-scope firm is registered when its data-year turnover is at or above T ; a voluntary registrant ($y_i < T_0$) is unchanged, since no threshold move changes its obligation; out-of-scope firms never remit. Total VAT revenue at threshold T is therefore

$$R(T) = \sum_i w_i v_i s_i \mathbf{1}\{y_i \geq T \vee r_i\},$$

which reproduces the baseline registered set at $T = T_0$. Three conventions follow and are worth stating. First, because voluntary registrants already remit, a threshold cut credits only the liability of in-scope firms newly drawn into the net: about 89% of below-threshold frame enterprises are already registered (Section 3), so the firms a cut mainly reaches are the unregistered stratum. Second, a threshold rise releases every in-scope registrant with data-year turnover in $[T_0, T)$: they no longer need to register, which is the reading HMRC’s costing uses; the statutory deregistration threshold (£2,000 below the registration threshold) would keep an existing registrant in the top of the band in the net, and I report that as a sensitivity. Third, the headline treats every released firm as deregistering; the anchor comparison reports the voluntary-retention sensitivity. Band membership is evaluated on *data-year* turnover in every fiscal year: a static cross-section has no growth process, and ageing the near-threshold stock across the threshold by a uniform factor would treat bunched firms as crossers, which they are not by construction (Section 5). Liabilities are aged by a cumulative nominal-growth factor—1.031, 1.052, 1.078, 1.110, and 1.142 for 2024–25 to 2028–29 relative to 2023–24, applied to the 2024–25 vintage relative to its own data year; the factors are assumptions. The estimate is the mechanical reclassification effect and excludes the behavioural response of Section 7. At the £90,000 threshold this puts the in-scope above-threshold VAT base at £184.1bn in 2025–26 and £188.7bn in 2026–27.²

Anchor comparison against HMRC’s published costing. I compare the static pipeline against HMRC’s costing of the April 2024 increase from £85,000 to £90,000, announced and costed at Spring Budget 2024 (HM Treasury, 2024). The comparison requires the costing baseline convention: the official pre-measure baseline is not a frozen £85,000 threshold but a counterfactual path in which the freeze legislated to March 2026 ends and the threshold resumes uprating (85, 85, 87, 89, 92 thousand pounds over 2024–25

²These aggregates are weighted sums of the in-scope registered population’s net VAT liabilities above the threshold, aged forward—a theoretical-liability (VTTL-style) base, not the published cash-receipts line (around £169bn in 2023–24; OBR forecast near £180bn for 2025–26), which is reduced by refunds, the VAT gap, and collection timing. Below-threshold voluntary remittances are excluded from the reported base because the model’s standard-rate-on-value-added liability is not calibrated for them (about £2.5bn model-implied against a £1.46bn HMRC total). The unaged bases are £174.7bn in 2023–24 and £180.5bn in 2024–25; the liability-by-band calibration scores (96.9% and 98.1%) summarise the remaining discrepancy against HMRC’s band totals. Band-to-band *differences*, which drive every costing in the paper, are less exposed than the level.

to 2028–29 in the model’s implementation), so the measure’s impact shrinks over the forecast and changes sign in 2028–29, when the counterfactual threshold overtakes £90,000. The official profile is $-150, -185, -125, -50, +65$ £m. Each year I difference revenue under the £90,000 policy and under that year’s counterfactual threshold, both evaluated with the registration rule above on data-year turnover and liabilities aged to the year. The mechanical headline, in which every released firm deregisters, gives $-193, -197, -120, -41, +88$: it reproduces the sign, the narrowing profile and the 2028–29 turning point, and lands within 7% of the official figure in 2025–26 and 4% in 2026–27, overshooting by 29% in 2024–25 (HMRC’s first year carries a registration lag the static rule has no counterpart for), undershooting by 18% in 2027–28, and overshooting the sign-flip year by 36% (where the result depends on the assumed £92,000 counterfactual threshold). No retention or behavioural parameter is tuned to produce this: the released population is the in-scope registrants with data-year turnover in the vacated band, about 28,000 firms in the first two years against HMRC’s expected 28,000 fewer registrants, and its liability is the calibrated standard rate on value added. Retaining the [Liu et al. \(2021\)](#) voluntary-registration share (43%) of released-firm liability gives $-110, -112, -68, -24, +88$, undershooting the official profile by 27–53%: LLAT’s share is a *firm* share among all below-threshold eligibles, and applying it as a liability share among just-released firms double-counts the voluntary registrants the rule already holds in the net, so it is a generous retention assumption rather than a central case. Keeping existing registrants above the £88,000 deregistration threshold in the net gives $-116, -119, -122, -41, +85$, a stock reading that does not follow the counterfactual path. Because my microdata are calibrated to HMRC VAT aggregates, even a close match confirms internal consistency rather than out-of-sample accuracy; HMRC’s own note expects about 28,000 fewer registrants in 2024–25 and 14,000 fewer on average over the forecast, against about 28,000 in-scope registrants in the released band here (Figure 4).

A menu of threshold locations. I next cost a menu of further threshold locations, taking the post-reform £90,000 threshold as the baseline. This sweep runs on the 2024–25 vintage population (Section 3). Each row of Table 1 costs a move of the threshold to a new location for 2025–26 by the registration rule above, holding the data-year turnover distribution fixed, so the revenue change reflects only the reclassification of in-scope firms into or out of the VAT net. Because the optimiser determines the within-band shape near the threshold (Section 3), and because the 2024–25 vintage has no fine-band targets, so that its density steps at the £90,000 HMRC band edge (Section 5), the $\pm$ £5,000 rows are conditional on that synthetic within-band allocation. Lowering the threshold draws in in-scope firms not already registered—mainly the unregistered stratum, whose liability under registration is the standard rate on its value added—while raising it releases registered firms in the vacated band. A £20,000 cut to £70,000 raises about £1.59bn and brings about 246,000 businesses into the net; a £30,000 rise to £120,000 costs about £1.28bn, 0.69% of the £184.1bn base. The cut rows credit the full standard-rate liability of newly covered businesses, with no allowance for the compliance cost or the registration lag a lower threshold would impose on them, so they are upper bounds on the first-year yield. Figure 5 shows the sweep graphically. These are *static* results: they bound the first-order

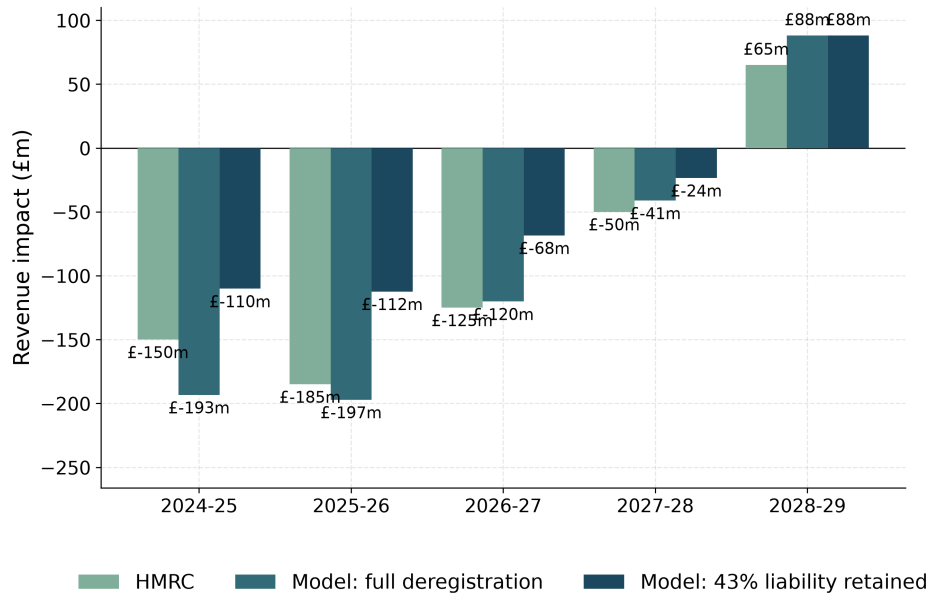


Figure 4: Anchor reform: revenue impact of raising the VAT registration threshold from £85,000 to £90,000 (the April 2024 change), my estimate under the full-deregistration and 43% retention conventions versus HMRC’s published costing, 2024–25 to 2028–29. Both series narrow over the forecast and turn positive by 2028–29 because the costing baseline is a counterfactual threshold path that resumes uprating after March 2026 and overtakes £90,000. The mechanical estimate, with no retention or behavioural parameter, lies within 30% of the official profile in every release year and within 10% in 2025–26 and 2026–27; the 43% retention convention undershoots it. Because the microdata are calibrated to HMRC and OBR aggregates, this is a consistency check, not an out-of-sample validation.

fiscal effect of a threshold change but abstract from the behavioural response of Section 7.

Table 1: Static revenue and VAT-paying-firm effects of moving the VAT registration threshold, 2025–26, relative to the post-reform £90,000 baseline (in-scope above-threshold VAT base £184.1bn; 2024–25 vintage population). Figures are mechanical, holding data-year turnover fixed (no behavioural response); in-scope firms only, including the unregistered stratum below the threshold. Voluntary registrants are held registered. Cut rows credit the full liability of newly covered businesses (see text). This sweep is distinct from the anchor reform of Figure 4. The machine-readable table is `results/static_sweep.txt` in the replication repository.

Threshold	Revenue change (£m)	Change (%)	VAT-paying firms (000s)
£70,000	+1,587.0	+0.86	+246.3
£75,000	+1,171.0	+0.64	+175.3
£80,000	+764.9	+0.42	+110.9
£85,000	+376.5	+0.20	+52.7
£90,000 (baseline)	0	0	0
£95,000	−198.0	−0.11	−28.8
£100,000	−404.4	−0.22	−57.4
£105,000	−625.9	−0.34	−86.3
£110,000	−845.6	−0.46	−113.7
£115,000	−1,066.0	−0.58	−139.6
£120,000	−1,278.7	−0.69	−163.2

4.1 Costing schedule reforms

The threshold sweep above moves the *location* of a single cutoff. The same static principle—re-applying a tax rule to the fixed turnover distribution, with no behavioural response—also costs reforms that change the *shape* of the schedule rather than merely its position. I consider four reforms and report their first-order revenue effects against a common baseline: the £85,000-notch / £174.7bn 2023–24 base of in-scope firms. (This 2023–24 data-year base differs from the £184.1bn figure in the sweep above, which runs on the 2024–25 vintage aged to 2025–26 at the post-reform £90,000 threshold; the schedule reforms are costed in the data year to match the bunching and dominated-region analysis. Both bases are theoretical-liability (VTTL-style) aggregates in the sense of the footnote above, not cash-receipts lines.) For each reform I recompute the weighted sum of liabilities $R = \sum_i w_i v_i$ under the new schedule, holding each firm’s turnover fixed, and difference it against the baseline.

Threshold relocation. Raising the threshold from £85,000 to £100,000 releases the firms with turnover in [$\$85,000, \$100,000$)—about 87,800 in-scope firms—from the VAT net, for a static revenue change of −£583m on the common base (the menu applies each schedule to in-scope firms at their data-year turnover, without the deregistration-gap

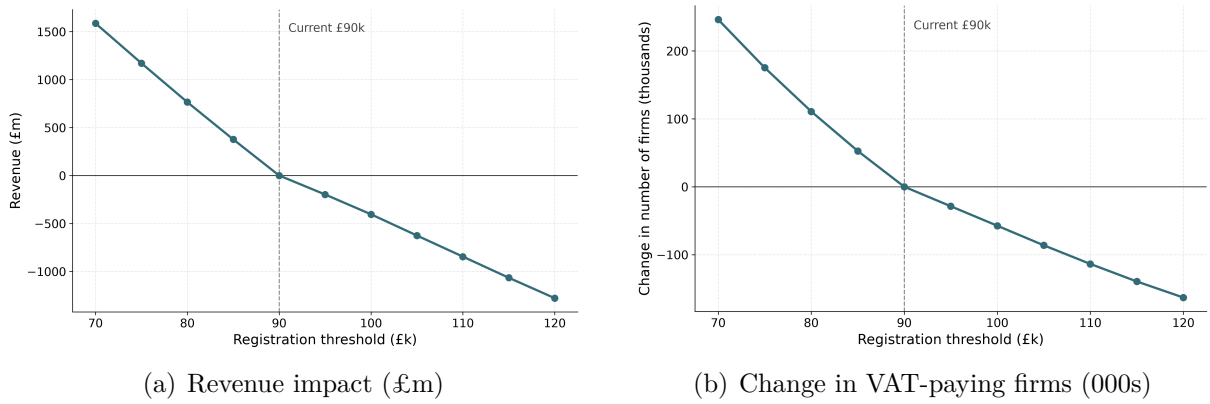


Figure 5: Static effect of moving the VAT registration threshold, 2025–26, relative to the £90,000 baseline (dashed line). Panel (a): revenue impact—cuts draw the unregistered stratum into the net; rises release registered firms. Panel (b): the change in the number of VAT-paying firms, tracking the local density of firms on each side of the threshold.

convention of the anchor). Because the released band lies *above* the data-year threshold, its liabilities are directly calibrated (and its lower reach pinned by the OBR fine-band targets), so the band is summed directly; a smooth-counterfactual alternative that fits the above-threshold per-£1,000 profile and integrates it across the band gives −£615m and 90,500 firms—within 6% of the direct sum now that the within-band fill is monotone. On this common base the level move relocates the dominated region to the new cutoff and widens it to £25,000 (Section 6) at a static cost of −£583m; the linear taper removes the dominated region entirely at the menu’s largest static cost of −£1,408m, and a constant-marginal-rate taper of the same width does so at −£932m (Section 6). Which matters more depends on what the reader wants from the reform—the dominated-region metric indexes misallocation only, and Section 8 discusses what it omits.

Graduated taper. A graduated taper replaces the discrete notch with a *marginal* remittance rate that phases in over a band, so liability is the integral of the marginal rate and a firm’s *effective* rate rises continuously from 0% to the full 20% across the band. The band top is a property of the chosen shape (Section 6): a taper whose marginal rate is constant at $m > \tau$ meets the standard-rate line at $U(m) = mT^*/(m - \tau)$, which is £141,667 at $m = 50\%$ and falls towards £106,250 as $m \rightarrow 100\%$, while a taper reaching the full rate at £105,000 would push net revenue inside the band below its value at the threshold, recreating a dominated interval smoothly rather than at a cliff. I cost two designs on the same £85,000–£141,667 band: the *linear* taper, whose marginal rate rises from 0% to 100% across the band, and a *constant* 50% marginal-rate taper. Firms in the band remit the tapered schedule on their (fixed) turnover; mechanically the linear design reduces revenue by −£1,408m and the constant-rate design by −£932m, with about 278,100 in-scope firms in the band facing a lower effective rate. The linear design gives more relief low in the band at the price of a marginal remittance rate that reaches 100% at the top; the constant-rate design caps the marginal rate at 50% and costs a third less.

Banded reduced rate. A banded reduced rate charges firms in the £85,000–£105,000 band a reduced rate $\tau_{\text{low}} \in \{10\%, 15\%\}$ in place of the standard 20%, with 20% retained above £105,000. Statically this reduces revenue by –£398m at $\tau_{\text{low}} = 10\%$ and by –£199m at $\tau_{\text{low}} = 15\%$, with about 117,300 in-scope firms in the band facing the lower rate.

Table 2: Static, behaviour-free revenue effect of the reforms relative to the common £85,000-notch / £174.7bn 2023–24 baseline of in-scope firms. Each reform is costed mechanically by re-applying its schedule to the fixed turnover distribution, with no behavioural response.

Reform	Lever	Static (£m)
Raise threshold to £100,000	Location	–583
Linear taper, £85k–£141.7k	Shape (marginal rate 0 → 100%)	–1,408
Constant 50% marginal taper, £85k–£141.7k	Shape (marginal rate 50%)	–932
Reduced rate 10%, £85k–£105k	Rate (step)	–398
Reduced rate 15%, £85k–£105k	Rate (step)	–199

These static figures bound the first-order fiscal effect of each reform but abstract from the behavioural response of Section 7. Two maintained conventions carry through every costing: the firm bears the full statutory rate on its value added (no pass-through to prices), and per-firm liability is the standard rate on value added, so Flat Rate Scheme users—who are concentrated in exactly the £85,000–£150,000 bands—are represented only through the calibrated band totals, not individually. One feature of the shape reforms is not visible in a static costing: unlike a threshold move, which transports the notch to a new cutoff, the taper and the reduced rate *change the size of the notch itself*—the discrete drop in net revenue at the threshold and the empty band of turnover it induces. Quantifying that change requires the structural object I develop in Section 6, where I derive the notch’s dominated region and show analytically how each of these reforms alters it.

5 Bunching and the limits of band-calibrated synthetic data

This section applies the polynomial counterfactual-density estimator of Chetty et al. (2011) and Kleven and Waseem (2013)—as applied to UK VAT data by Liu et al. (2021)—to the synthetic firm population. Its finding is about the data, not about firm behaviour: every sub-band feature of a band-calibrated synthetic population is exactly what its targets put there, and this section demonstrates the point using the 2023–24 population. Its near-threshold density is calibrated to the OBR’s published £1,000-band profile (Section 3), and the estimator reads that profile back as excess mass—target inheritance, not behaviour. The results establish the section’s transparency claim: band-calibrated

synthetic data support no bunching inference in either direction, and any sub-band feature of such data should be traced to its calibration targets before it is read as economics. The behavioural evidence on UK VAT bunching remains the administrative estimate of [Liu et al. \(2021\)](#).

The section also records a correction, because it is the sharpest available demonstration of the thesis. An earlier draft reported an excess mass of 8,712 firms at £85,000 that appeared to reproduce the administrative density step. It was an artifact of a liability mis-scaling in the generator (net liability set to value added rather than the standard rate on value added), which distorted how the weight optimiser distributed mass near the threshold; a later intermediate build produced a still larger, specification-robust artifact at the £90,000 band edge from a mis-specified below-threshold liability target. Aggregate calibration scores barely moved under either defect, which is the point: aggregate fit does not discipline sub-band shape. The replication repository’s changelog records each correction and its diagnosis.

5.1 Methodology

Following [Liu et al. \(2021\)](#), I group firms into fine turnover bins of £1,000 and estimate a counterfactual density—the distribution that would obtain absent the registration notch—by polynomial regression on bin counts, excluding a manipulation window around the threshold y^* :

$$c_j = \sum_{l=0}^q \beta_l (y_j)^l + \sum_{i \in [y_-^*, y_+^*]} \gamma_i \mathbb{1}\{j = i\} + \varepsilon_j,$$

where c_j is the number of firms in bin j , y_j is the distance of bin j from the threshold, q is the polynomial order, and $[y_-^*, y_+^*]$ is the excluded window. The fitted polynomial is then projected into the excluded region to recover the counterfactual density $g(y)$. The upper edge of the manipulation window, y_R , is then searched for by walking upward from the threshold until cumulative missing mass equals the excess mass below; when the missing mass inside the search window is insufficient—as in the headline specification, where the fitted missing mass is far short of the excess— y_R is censored at the window edge and reported as such (Appendix A.2). Excess mass is the integrated positive part of the gap between the observed density f and the counterfactual g in the excluded window below the threshold,

$$E = \int_{y^*-W}^{y^*} [f(y) - g(y)]_+ dy,$$

a gross rather than net statistic. The normalised excess b_{LLAT} divides the integrated positive-part gap over the whole manipulation region $[y^* - W, y_R]$ by the average counterfactual height over that region—a literature-style normalisation in the spirit of [Liu et al. \(2021\)](#), who report excess mass relative to the counterfactual density, though it does not reproduce their estimator exactly. Because it also counts positive gaps above the threshold, b_{LLAT} is not E divided by a density, and the two statistics need not move together. Neither is an elasticity. The estimator delivers the raw geometry of any density

step at the threshold; whether that step reflects firm behaviour is a separate question. The estimator reports geometry only (b_{LLAT} , excess mass E , and the mass-conservation objects): no elasticity is computed from it, both because the identification critique of bunching designs applies (Blomquist et al., 2021; Bertanha et al., 2023) and because on these data there is no behavioural mass to convert.

5.2 Results: diagnostics, not behavioural estimates

At the £85,000 threshold (Figure 6) the baseline specification returns $E = 8,268$ weighted firms and $b_{\text{LLAT}} = 1.411$ on the all-business density (frame plus unregistered stratum, the universe of the OBR chart). This reads back a profile calibrated to the OBR fine-band levels; the generator contains no firm location choice, so it is not a behavioural estimate. Across the degree-window grid, E ranges from about 560 to about 157,600 firms and the headline fitted missing mass above the threshold is small ($\Delta_R = 896$), with y_R censored at the search cap (Appendix A.2). The 2024–25 population, which has no fine-band targets, makes the complementary point: at its £90,000 threshold the same estimator returns $E = 2,475$ and $b_{\text{LLAT}} = 1.445$, an excess mass produced entirely by the step between the two HMRC coarse bands that meet at £90,000, which the weight optimiser resolves as a density step. Neither figure is evidence about firm responses; both are audits of what the estimator reads from the synthetic construction, and the second is the reason the $\pm\$5,000$ rows of Table 1 are flagged as conditional on the within-band allocation. Two constructed-data exercises in the replication repository characterise the estimator itself. On a placebo population whose density is reweighted to a smooth profile across £85,000, the estimator returns $E = 0$: it produces no false positive on a step-free world. When a Kleven–Waseem-consistent relocation of known size is injected into that placebo, the estimator recovers 0%, 11% and 30% of injected excess masses of 2,000, 5,000 and 8,000 firms, with mass conservation then binding at an interior y_R : the counterfactual is fitted on bins that include the depressed region beyond the exclusion window, so the estimator attenuates a genuine response rather than inflating it (`results/placebo_bunching.txt`, `results/recovery_bunching.txt`).

5.3 Comparison with the literature

The behavioural fact in the literature is administrative. Liu et al. (2021) (hereafter LLAT) document genuine bunching at the UK VAT registration notch in HMRC micro-data over 2004–05 to 2014–15, during which the threshold rose from £58,000 to £81,000, with an excess-bunching ratio of $b = 1.361$ (SE 0.202) around the year-specific threshold and a manipulation window of $-\$14,000$ to $+\$24,000$. The synthetic baseline’s $b_{\text{LLAT}} = 1.411$ is not a second estimate of their parameter: it is an unstable readback of the OBR target profile. What does carry across is institutional context: LLAT report that almost half (roughly 43%) of eligible below-threshold firms register voluntarily, which dampens observed bunching relative to a pure notch and which motivates the voluntary-retention sensitivity of Section 4. For wider order-of-magnitude context, Office for Budget Responsibility (2023) count firms holding turnover below the frozen threshold over a *wide*

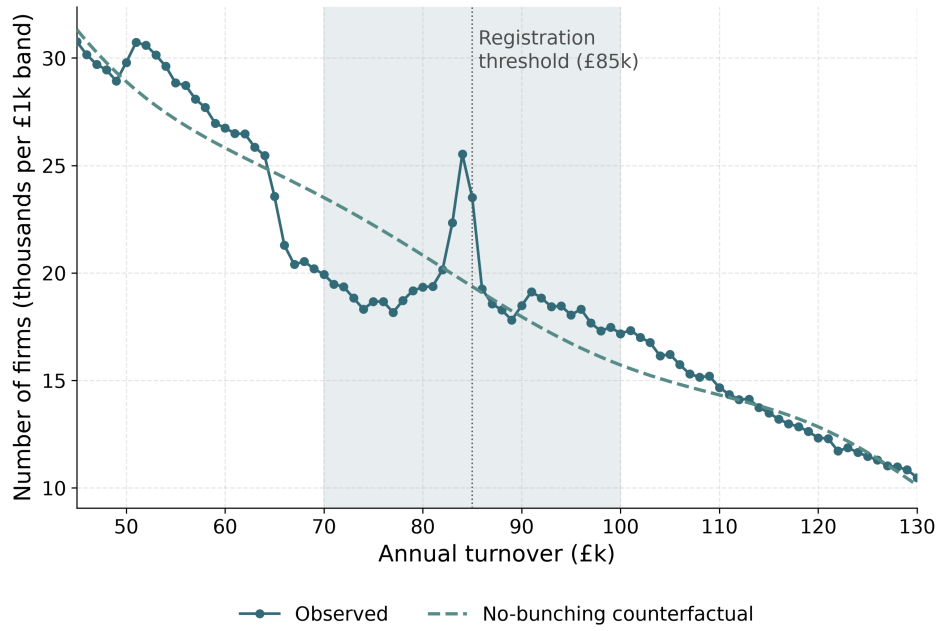


Figure 6: Observed synthetic firm density and polynomial no-bunching counterfactual at the 2023–24 registration threshold. The 2023–24 population reproduces the administratively observed bunching profile because it is calibrated to the OBR £1,000-band targets; the estimator reads the target-inherited excess back as $E = 8,268$ ($b_{\text{LLAT}} = 1.411$). This is not behavioural: the generator has no firm location-choice, so all sub-band structure is inherited from the targets and the weight optimiser.

band (rising from 23,000 in 2017–18 to about 44,000 in 2025–26, with foregone turnover of £110m rising to £350m); this is a different object from a near-threshold spike and is cited only as context.

5.4 Reduced-form evidence and the dominated region

These results sharpen, rather than weaken, the case for grounding the policy analysis in structure. The reduced-form density around the threshold cannot price reforms—it has no single point of excess mass to forward-map onto a change in the *shape* of the schedule—and on band-calibrated synthetic data it cannot be read behaviourally at all, in either direction. What the structure of Section 6 *does* deliver exactly, for every reform considered, is the notch’s dominated region and how each schedule change alters it. It is precisely because the reduced-form bunching is non-identifying here that the exact dominated region carries the distortion characterisation. The behavioural layer that re-prices each firm’s turnover under counterfactual schedules—conditional on an assumed turnover elasticity—is developed in Section 7.

6 The notch and its dominated region

The behavioural response to the registration notch is documented in administrative data by Liu et al. (2021); the synthetic data contain no bunching signal to exploit (Section 5). To characterise the notch precisely I therefore turn to its analytic structure, which yields an exact distortion measure that needs no behavioural calibration. I set out the firm-level structure following Kleven and Waseem (2013), Saez (2010), and Kleven (2016), whose bunching framework Liu et al. (2021) adapt to UK VAT. The UK registration threshold is a *notch*, not a kink: once turnover crosses the threshold the firm becomes liable for VAT on its *entire* base, not merely on the increment above it. The section’s central deliverable is the notch’s *dominated region*—an exact, elasticity-free property of the schedule that indexes the distortion the threshold imposes (Section 6.2)—and its analytic response to each schedule reform, complementing the static reform costs of Section 4.1. Section 7 prices the intensive-margin response to each reform conditional on an assumed turnover elasticity; nothing in this section depends on that assumption.

Scope. The model prices the registration margin for firms that remit positive net VAT. Two populations are outside its scope and are treated as such throughout: below-threshold *voluntary* registrants (roughly 43% of eligible firms in Liu et al., 2021), who have chosen registration and so face no notch at the threshold, and net-repayment traders (zero-rated exporters and similar), whose remittance is negative for rated-structure reasons the generator does not represent (Section 3).

6.1 The firm’s problem

Each firm i chooses turnover y_i to maximise after-tax profit. Let $c_i(y_i)$ denote the cost of producing turnover y_i , T^* the registration threshold, and τ the standard VAT rate. In the

simplest statement VAT applies to the firm's entire turnover once it registers, so after-tax revenue is discontinuous and profit is

$$\pi_i(y_i) = \begin{cases} y_i - c_i(y_i), & y_i < T^* \quad (\text{unregistered}), \\ (1 - \tau) y_i - c_i(y_i), & y_i \geq T^* \quad (\text{registered}). \end{cases} \quad (1)$$

At T^* after-tax revenue falls discretely by τT^* : a firm that nudges its turnover from just below to just above the threshold loses τT^* of net revenue on turnover it was already earning. With the standard rate $\tau = 0.20$ and threshold $T^* = \$85,000$ this discrete drop is $\tau T^* = \$17,000$ on whole turnover. Real VAT taxes value added, and the corrected data give each firm net liability $v_i = \tau(y_i - x_i)$ with deductible inputs $x_i = \delta_i y_i$ (Section 3); profit is then $(1 - \delta_i)y_i - \tilde{c}_i(y_i)$ unregistered and $(1 - \delta_i)(1 - \tau)y_i - \tilde{c}_i(y_i)$ registered—the deductible share multiplies *both* branches, so the notch geometry of (1) carries over firm by firm, with the cash size of the jump scaled to $\tau(1 - \delta_i)T^*$ (about £6,800 at the mean value-added share of 40%). This drop—absent from a kink, where only the *marginal* rate changes at the threshold while average liability moves continuously—is the defining feature of a notch: it makes locating just below the cutoff strictly attractive and creates the empty band derived next. Throughout, the firm is assumed to bear the full statutory rate on its value added—no pass-through to prices—so the notch geometry is stated in terms of the firm's own net revenue; with partial pass-through the cash size of the jump, and the width derived next, would shrink in proportion to the share borne by the firm.

6.2 The dominated region

The notch creates a range of turnover just above T^* that no profit-maximising firm chooses under the model. A firm just below the threshold earns net revenue T^* ; a registered firm at $T^* + a$ earns $(1 - \tau)(T^* + a)$. Before these net revenues become equal, the registered firm receives less net revenue while producing more. Thus, for any non-decreasing production cost, it earns strictly lower profit; no locally constant-cost approximation is required. Equating net revenues defines the boundary of this guaranteed dominated region:

$$(1 - \tau)(T^* + a) = T^* \implies a = T^* \frac{\tau}{1 - \tau}. \quad (2)$$

This is the Kleven–Waseem guaranteed dominated-region width (Kleven and Waseem, 2013). At my parameters it gives $a = \$85,000 \times 0.20/0.80 = \$21,250$, so the dominated region is $(T^*, T^* + a) = (\$85,000, \$106,250)$, shown in Figure 7.

Two properties make this band a transparent index of the distortion. First, it is *exact and elasticity-free*, and—less obviously—*invariant to the firm's input share*. Under the value-added formulation above, the indifference condition at the edge of the region is $(1 - \delta_i)(T^*) = (1 - \delta_i)(1 - \tau)(T^* + a)$: the deductible share $(1 - \delta_i)$ cancels, so every firm with positive value added faces the *same* dominated width $a = T^* \tau / (1 - \tau)$, whatever its input intensity. (A per-firm width built from the net remittance rate, such as $T^* \tau_{0i} / (1 - \tau_{0i})$, corresponds to a different technology in which marginal turnover carries no input cost,

and is not used here.) The width depends on τ and T^* alone—not on the cost specification beyond monotonicity, any behavioural elasticity, or any smoothing parameter. Who escapes the notch is a matter of scope, not of input shares (the invariance is a property of the formulation: unregistered profit carries the same deductible share, so no input VAT goes unreclaimed below the threshold; if deregistered firms instead bore VAT on their inputs, the width would shrink in the input share and vanish for shares above one half—the input-reclaim margin that drives voluntary registration, Section 2): voluntary registrants escape by choice, and net-repayment traders have no positive remittance to jump (the Scope paragraph above). Second, the region is an order of magnitude wider than the $\sim \$1k$ region implied by a smoothed *kink* approximation, and it spans a populous part of the firm distribution: about 124,300 weighted in-scope firms in the calibrated synthetic population lie in the \$85,000–\$106,250 band (`results/dominated_region_mass.txt`; out-of-scope enterprises face no notch and are excluded). Because the generator has no location choice, this counts the firms the data *place* in the band—indexing the size of the population exposed to the distortion—rather than firms that optimally locate there; the count is conditional on the uniform within-band fill inside the coarse ONS bands (Section 3) and would shift under a different within-band shape. It is the width a , not the count, that is exact.

How each reform changes the dominated region. Because the width $a = T^*\tau/(1-\tau)$ depends on the threshold and the rate alone, the effect of each schedule reform of Section 4.1 on the dominated region is exact and follows directly from the notch arithmetic of (2) and (3), with no dependence on the cost specification beyond monotonicity, any behavioural elasticity, or any simulated re-optimisation. Raising the threshold to £100,000 leaves the rate unchanged and therefore *relocates* the region—and, because the width is proportional to the threshold, *widens* it: at $T^* = \$100,000$, $a = \$100,000 \times 0.20/0.80 = \$25,000$, so the dominated region becomes (\$100,000, \$125,000). Only the proportional width $a/T^* = 25\%$ is unchanged; the absolute band grows by £3,750. A banded reduced rate does *not* remove the distortion. It reduces the *largest single* notch—the jump at T^* falls from $\tau T^* = \$17,000$ to $r T^* = \$12,750$ at a 15% band rate—but, because the band reverts to the standard rate τ at its upper edge $T_1 = \$105,000$, it introduces a *secondary* notch there, and hence a second dominated region the simple “ a shrinks with τ ” reasoning omits. The primary region at T^* (the band rate r on whole turnover) has the usual width $a = T^*r/(1-r)$: $\$85,000 \times 0.15/0.85 = \$15,000$ at 15% and $\$85,000 \times 0.10/0.90 = \$9,444$ at 10%. The secondary region at the band top—where the schedule steps from r back to τ on whole turnover—has width

$$a' = T_1 \frac{\tau - r}{1 - \tau}, \quad (3)$$

giving $\$105,000 \times 0.05/0.80 = \$6,562$ at 15% and $\$105,000 \times 0.10/0.80 = \$13,125$ at 10%. The *total* dominated-turnover width is therefore $a + a' = \$21,563$ at 15% and $\$22,569$ at 10%—both marginally *above* the \$21,250 baseline, not below it. The reduced rate splits one wide empty band into two narrower ones rather than shrinking the total: it lowers the largest single notch (to \$12,750 at 15%) at the price of adding a secondary notch of $(\tau - r)T_1 = \$5,250$. The firm counts confirm this (`results/dominated_region_mass.txt`):

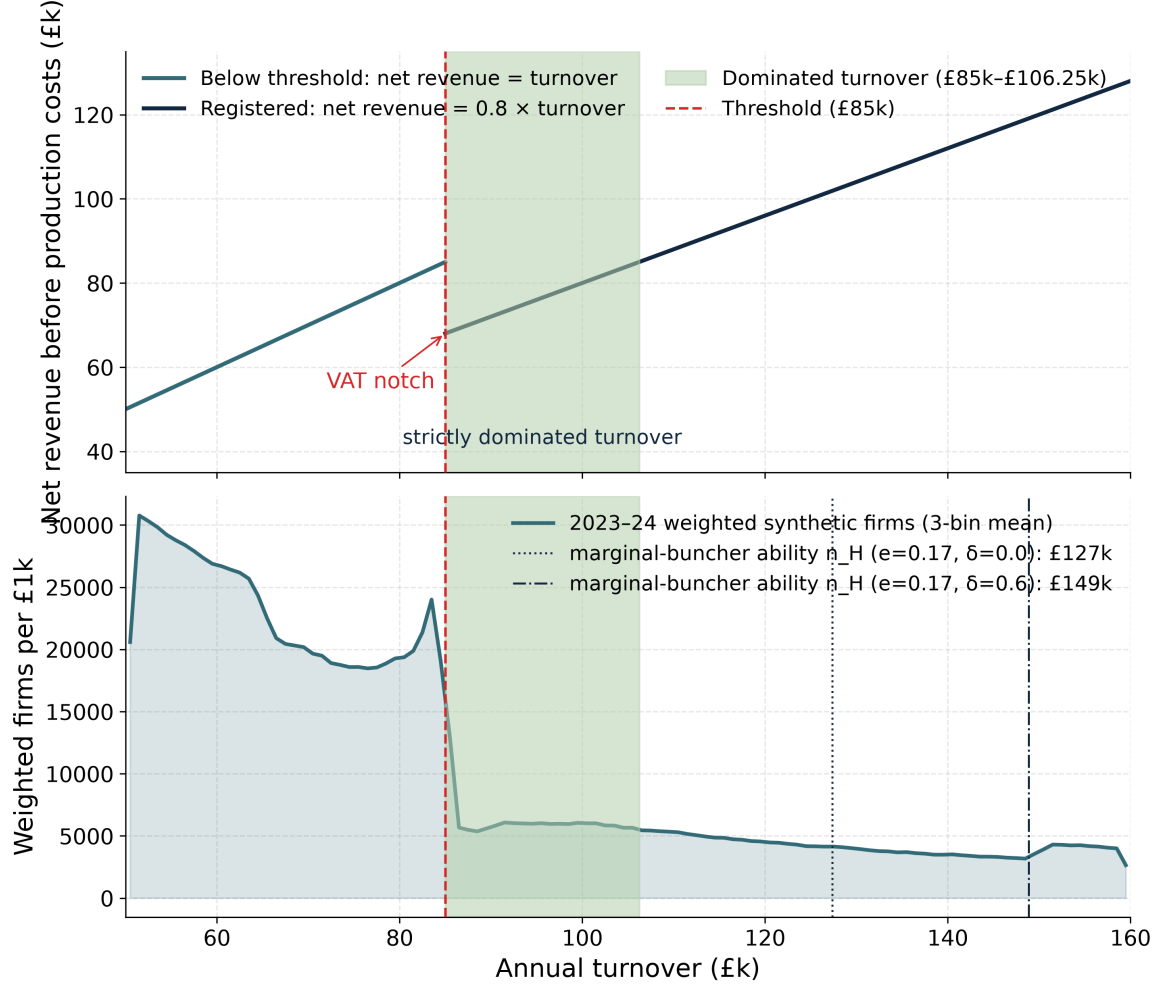


Figure 7: The VAT notch and the firm distribution. Panel (a) shows net revenue before production costs: immediately above £85,000, VAT reduces it from y to $0.8y$, making turnover in $(\$85,000, \$106,250)$ strictly dominated. Panel (b) shows the three-bin moving mean of weighted 2023–24 in-scope synthetic firm counts, with the same interval shaded. The distribution is descriptive and target-inherited, not the outcome of behavioural optimisation; its step at £100,000 reflects the boundary between coarse ONS source bands, not an empty turnover interval. The vertical lines mark the marginal-buncher *ability* n_H at $e = 0.17$ for deductible shares $\delta = 0$ and $\delta = 0.6$ (Section 7); a firm of that ability registers at $n_H[(1 - \delta)(1 - \tau)]^e$, so the lines are not observed-turnover locations. The exact dominated width is $a = \$21,250$.

the primary regions hold about 87,800 (15%) and 54,500 (10%) weighted in-scope firms, the secondary regions add about 35,300 and 67,200, and the totals—123,100 and 121,700—lie within 2% of the 124,300-firm baseline band. The total dominated *width* is larger under both reduced rates and the exposed population is essentially unchanged: the reduced rate relocates the distortion, it does not remove it. A graduated taper, by contrast, can remove the dominated region entirely—provided its schedule keeps net revenue monotone, and that condition is restrictive. Any band-confined schedule whose effective rate reaches the full τ by £105,000 leaves net revenue $(1 - \tau) \times \$105,000 = \$84,000$ at the band top, below the £85,000 available at the threshold itself, so net revenue must fall over some range inside the band—a smoothly dominated interval in place of the cliff. Monotonicity requires the marginal remittance rate $m(y)$ to stay below one, and band confinement requires liability to meet the standard-rate line τy at the band top U . For a constant marginal rate $m > \tau$ these two conditions pin $U(m) = mT^*/(m - \tau)$: the band top is a property of the schedule’s *shape*, falling from £141,667 at $m = 50\%$ towards the infimum $T^*/(1 - \tau) = \$106,250$ as $m \rightarrow 100\%$, where net revenue is flat across the band and the region is only weakly dominated. The linear taper costed in Section 4.1 phases m from 0 to 100% over $[\$85,000, T^*/(1 - 2\tau)] = [\$85,000, \$141,667]$; the constant-rate taper costed alongside it holds $m = 50\%$ over the same band. Both leave liability continuous at both band edges and net revenue non-decreasing throughout, so the band of dominated turnover, with the discrete incentive to suppress turnover, is removed entirely. The price of removal is a wide band and, for the linear design, a marginal remittance rate that reaches 100% at its top; the constant-rate design caps the marginal rate at 50% and costs a third less. On the dominated-region metric, then, only the taper removes the distortion; the reduced rate splits it and the level move relocates and widens it. The metric indexes the misallocation distortion only—what it omits, and how the reforms compare on cost, is taken up in Section 8. The dominated region is the paper’s exact, elasticity-free measure of the distortion the notch imposes, and the analytic statement above is exactly how much of that distortion each reform of Section 4.1 removes.

7 Behavioural reform costs

The reforms of Section 4 are costed *statically*: each schedule is re-applied to a fixed turnover distribution, and the revenue change is the mechanical reclassification of firms with no response in turnover. Section 5 shows that the synthetic data contain no behavioural signal from which a response could be estimated, and Section 6 supplies the one object that needs no behavioural calibration at all: the exact dominated region. What is missing from this sequence is the behavioural layer itself—a costing that lets firms re-optimize turnover when a reform changes the effective rate they face. This section supplies that layer for an *assumed* turnover elasticity e , reported as an e -sensitivity range rather than a point estimate, and nesting the static costing exactly in its $e \rightarrow 0$ limit. Its headline is deflationary, and usefully so: on the intensive margin the static costings are robust. A pure threshold rise has *exactly* zero intensive-margin revenue offset at every e , and the reduced-rate bands’ offsets stay under 6% of the static cost across the sweep.

7.1 An iso-elastic simulator on the intensive margin

I adapt the iso-elastic model of Kleven and Waseem (2013), in the sufficient-statistics tradition of Saez (2010), writing the tax on *value added* rather than on turnover. (Liu et al. (2021) caution that turnover-tax bunching formulae do not carry to VAT unmodified, because input costs intervene; the value-added statement below is the adaptation this requires.) In this statement—formulation A—a firm of ability n_i chooses turnover y to maximise

$$\pi_i(y) = (1 - \delta_i)(1 - \tau f(y))y - C(y; n_i, e), \quad C(y; n, e) = \frac{n}{1 + 1/e} \left(\frac{y}{n}\right)^{1+1/e}, \quad (4)$$

where $\delta_i \in [0, 1]$ is the firm’s *deductible-input share* of turnover, so that value added is $(1 - \delta_i)y$; τ is the statutory VAT rate; and $f(y) \in [0, 1]$ is the fraction of τ that the schedule levies on the firm once registered ($f = 0$ below the threshold, $f = 1$ for a fully registered firm under the standard rate). Here $e > 0$ is the elasticity of turnover with respect to the net-of-tax rate.

It is essential to read C correctly. Because the deductible inputs $\delta_i y$ are already netted out of revenue through the $(1 - \delta_i)$ factor, $C(y; n_i, e)$ must be understood as the firm’s *own-factor production cost*—the labour and capital effort the firm itself supplies—and *not* its total cost of production. Reading C as total cost would double-count the bought-in inputs, which the value-added factor has already removed. Marginal own-factor cost is $(y/n)^{1/e}$, so with no tax and no bought-in inputs ($\delta = 0$) the optimum is $y = n$; with a positive deductible share the untaxed optimum is $n(1 - \delta)^e$, and ability n is the firm’s frictionless turnover in the own-factor technology. A firm registered under a flat rate ($f \equiv 1$) chooses

$$y^* = n [(1 - \delta)(1 - \tau)]^e, \quad (5)$$

so the retained wedge facing the firm is the product $(1 - \delta)(1 - \tau)$ of the value-added share and the statutory net-of-tax rate, and $d \ln y^* / d \ln [(1 - \delta)(1 - \tau)] = e$: e is exactly the knob that governs the intensive response. Figure 8 traces the formulation-A profit and its optima across the notch and taper schedules for three deductible shares.

The response ratio is free of the deductible share. The reform costing needs each firm’s optimal turnover under a counterfactual schedule, *given* that its observed turnover is optimal under the baseline notch. Formulation A makes this a one-line object. Inverting (5) at the baseline (fraction f_0) and re-solving at the reform (fraction f_1 on the firm’s region of the schedule), the ability and the deductible share cancel in the ratio:

$$y_i^* = y_{\text{obs},i} \left(\frac{1 - \tau f_1}{1 - \tau f_0} \right)^e. \quad (6)$$

The intensive response depends only on e , the statutory rate, and the schedule fractions—not on δ_i , not on the firm’s net remittance rate, and not on the recovered ability level. Ability recovery ($n_i = y_{\text{obs},i} / (1 - \delta_i)^e$ below the threshold, $n_i = y_{\text{obs},i} / [(1 - \delta_i)(1 - \tau)]^e$ above; the released code sets $\delta_i = 0$ in this anchor, which is immaterial because no reform number depends on it) is an *accounting anchor*—it rationalises the observed allocation given e

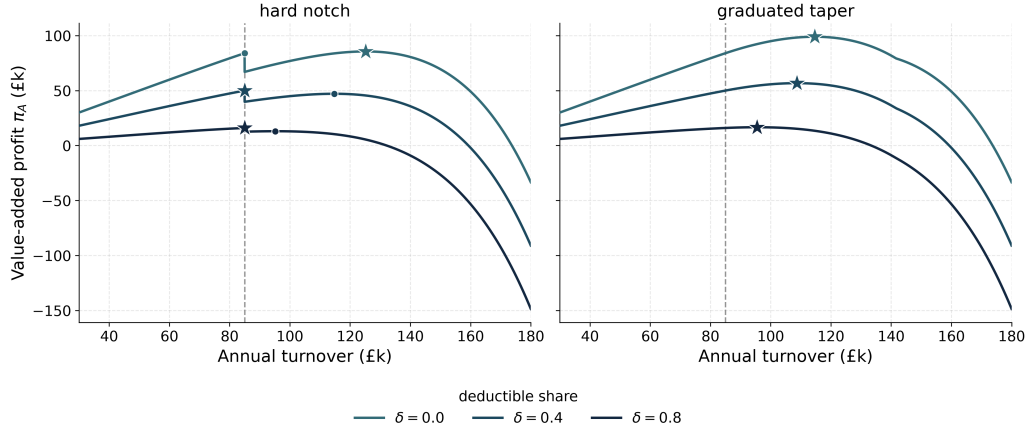


Figure 8: Optima of the formulation-A profit function (4) at $\tau = 0.20$, $n = \$130k$, $e = 0.17$, under the registration notch (left) and the graduated taper (right), for deductible-input shares $\delta \in \{0.0, 0.4, 0.8\}$. Dots mark local maxima, the star the global optimum, and the dashed line the £85,000 threshold. A higher δ scales the profit curve down and pulls the registered optimum $y^* = n[(1 - \delta)(1 - \tau)]^e$ toward the threshold, eventually tipping the global optimum to bunching at the notch.

and reproduces the baseline by construction—but no reform number below depends on it.

Region-confined scope. The margin priced here is purely *intensive*: each firm re-optimises within the region of the reform schedule that contains its observed turnover, per (6), clipped at its region’s edges. Relocation across a notch—bunching below a threshold from above, or de-registering past a band edge—is the *extensive* margin, which is the separate analytic object of Section 6 (the dominated region and the marginal buncher) and is deliberately out of scope: the synthetic data contain no information about it (Section 5), and pricing it would require assumptions about re-bunching dynamics the paper has declined to make. Confining the response to the firm’s own region also makes the $e \rightarrow 0$ limit collapse onto the static costing *exactly*, with no solver tolerance.³

Conditional on e . The elasticity e is not identified from these data, so every behavioural magnitude below is conditional on the assumed e , and the sweep is a sensitivity analysis rather than a set of forecasts. The sweep values $e \in \{0.05, 0.17, 0.32\}$ are assumptions: 0.05 is the external anchor—Kleven and Waseem (2013) find that structural elasticities recovered from notches, once optimisation frictions are accounted for, are small—and 0.17 and 0.32 are chosen to span the range used in the threshold literature. Liu et al. (2021) deliberately do not convert their bunching estimate into a turnover elasticity, because the Kleven–Waseem formula presumes business-to-consumer sales, price-taking, and a fixed-coefficient technology, so no published UK VAT-base elasticity anchors the

³An earlier version solved the registered optimum by damped fixed-point iteration, which has no fixed point for abilities that straddle a reform notch; the replication repository’s changelog records the change to the closed-form region-confined solve.

sweep. A reader may substitute any preferred value directly. The robust, e -free headline objects remain the static costs of Section 4 and the dominated region of Section 6.

7.2 Results

Table 3 reports the behavioural cost of each of the three flat-rate reforms—the raised threshold and the two reduced-rate bands—measured as the change in revenue relative to the £85,000-notch baseline, alongside the static figure of Table 2, for each e . For context, the marginal buncher $n_H(e, \delta)$ —the highest ability that still finds bunching at the threshold optimal, the extensive-margin object held fixed here—depends on the deductible share, because δ scales revenue but not own-factor cost in formulation A: with $\delta = 0$ it sits at £112,795, £127,382 and £143,527 for $e = 0.05, 0.17, 0.32$; at the population-mean input share $\delta = 0.6$ it rises to about £118,100, £148,900 and £192,400 (`results/dynamic_reform_results.txt`). Firms with higher input shares find registration less attractive relative to bunching, as Figure 8 shows.

Table 3: Behavioural revenue effect of the three flat-rate reforms relative to the £85,000-notch baseline, by assumed turnover elasticity e , with the static ($e \rightarrow 0$ limit) figure for comparison. Each behavioural figure re-solves every firm’s optimum within its schedule region per (6), holding e fixed. The final column reports the number of firms whose optimal turnover moves by more than £500 at $e = 0.17$. Estimates are conditional on the assumed e (Section 7.1). The graduated taper is excluded; see the paragraph below.

Reform	Static	$e = 0.05$	$e = 0.17$	$e = 0.32$	Firms ($e=0.17$)
Raise threshold to £100,000	–\$583m	–\$583m	–\$583m	–\$583m	84,893
Reduced rate 10% [\$85k, \$105k]	–\$398m	–\$395m	–\$390m	–\$384m	114,336
Reduced rate 15% [\$85k, \$105k]	–\$199m	–\$197m	–\$193m	–\$188m	114,336

Note. The tapers are omitted: their rate varies continuously with turnover, so their intensive response is outside the region-confined solve, and their marginal-rate channel is not represented by the flat-rate first-order condition. Their e -free figures—the static costs of –£1,408m (linear) and –£932m (constant 50%) and the exact removal of the dominated region—are in Sections 4 and 6. Firms are in-scope VAT firms with turnover in £70,000–£130,000, the range the region-confined solve re-optimises.

A level rise has no intensive-margin offset. The raise-to- £100,000 row is identical to its static cost at every e ; this follows from the region-confined convention rather than from a failure to respond. Partition the firms. Those with observed turnover below £85,000 face $f = 0$ before and after; nothing changes. Those released by the raise ($y_{\text{obs}} \in [\$85\text{k}, \$100\text{k})$) see their rate fall to zero and expand per (6) by the factor $(1 - \tau)^{-e}$ (about +3.9% at $e = 0.17$)—but their expansion is *untaxed*: they have left the VAT base, so the revenue loss is their full baseline remittance, exactly the static figure. Those above £100,000 face $f = 1$ before and after; nothing changes. The only candidates for a taxed response are released firms whose expansion would carry them across the new threshold. The region-confined solve does not evaluate that discrete choice: it clips such firms at the new threshold by construction, so the invariance is a property of the maintained status

convention—released firms deregister and are not permitted to re-cross the new notch—rather than a computed statement about their global optimum. Under that convention, base broadening from a threshold rise is real economic activity that generates no VAT, and the static costing of a level move coincides with the intensive-margin behavioural costing. A released firm whose ability exceeded the new marginal buncher would instead register at its taxed optimum and remit VAT; that is an extensive-margin case outside this exercise. The simulator asserts the conditional invariance programmatically (to £0.1m at each swept e).

Reduced-rate offsets are second order. A reduced-rate band lowers the rate on band firms from τ to r , so each scales up by (6) with $f_1 = r/\tau$: about +2.0% at the 10% rate and $e = 0.17$, and about +1.0% at 15%. Remittance scales with turnover, so the offset is roughly the scale gain times the band’s (rate-reduced) remittance—about £8m against a static £398m at $e = 0.17$ for the 10% band, reaching about £14m only at $e = 0.32$. The direction is uniform—a larger e makes the bands slightly cheaper—but the magnitude never approaches the static term. Figure 9 traces the costs across the full range of e : the raise is exactly flat, the bands nearly so.

Validation. Three properties are asserted programmatically in the released code on every run: the baseline solve reproduces each firm’s observed turnover exactly (the accounting anchor is internally consistent); the $e \rightarrow 0$ behavioural cost equals the static cost to £0.1m for every priced reform; and the solved response satisfies $d \ln y^* / d \ln(1 - \tau f) = e$ to machine precision. The static costing is thus the $e = 0$ corner of the same machine, not a separate calculation.

This section completes the costing framework, and its contribution is a robustness statement. Where Section 4 prices each reform mechanically and Section 6 measures the distortion each removes, the intensive-margin layer shows that naming an elasticity barely moves the static numbers: not at all for a level move, and by under 6% for the reduced-rate bands. The behavioural uncertainty that could move these costings materially sits on the *extensive* margin—registration, de-registration, and location choice at the notch—which the synthetic data cannot identify and which this paper prices only through its exact analytic geometry.

8 Conclusion

This paper contributes an open, auditable firm-level accounting microsimulation for analysing the UK VAT registration threshold, released in full—including its synthetic-data generator—for replication. Its value is as transparent policy-scenario infrastructure rather than a structural identification result: I make no claim to recover policy-invariant firm primitives, unlike Garicano et al. (2016) and Gourio and Roys (2014). The analysis is anchored to the 2023–24 data year, in which the statutory threshold was £85,000, with a second 2024–25 population (threshold £90,000) for the forward sweep. Table 4 collects the headline objects.

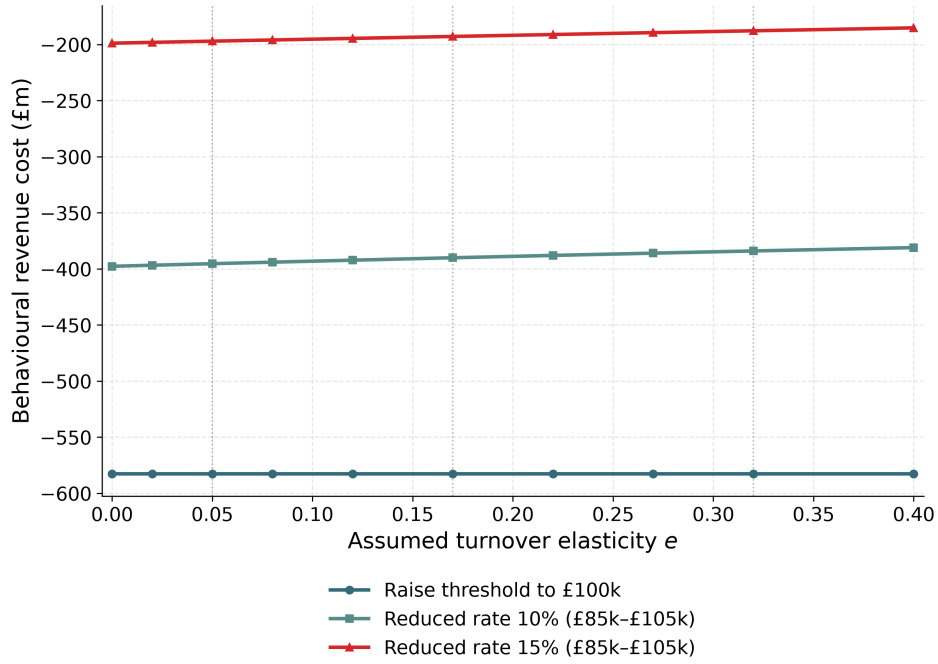


Figure 9: Behavioural reform cost against the assumed turnover elasticity e , for the three flat-rate reforms. Each curve is the change in revenue relative to the £85,000-notch baseline; the intercept at $e = 0$ is the static cost of Table 2. The raise-to-£100,000 line is exactly flat (no intensive-margin offset); the reduced-rate lines decline by under 6% of their static values across the plotted range. Dotted lines mark the swept values $e \in \{0.05, 0.17, 0.32\}$. Estimates are conditional on the assumed e (Section 7.1).

Table 4: Summary of headline estimates, consolidating results defined in the body sections cross-referenced below. Rows 1–2 are 2025–26 figures (the anchor on the 2023–24 population, the sweep on the 2024–25 population at its £90,000 baseline); the remaining rows refer to the £85,000 data-year baseline.

Object	Definition	Value	Source
Anchor, £85k to £90k, 2025–26	mechanical / 43% retention	–\$197m / –\$112m (HMRC –\$185m)	§4
Sweep, £90k baseline	£70k to £120k	+\$1.59bn to –\$1.28bn	Table 1
Dominated region	$a = T^*\tau/(1 - \tau)$	£21,250	§6.2
In-scope firms in dominated band	weighted	$\approx 124,000$	§6.2
Raise to £100k	static = behavioural	–\$583m	§4, §7
Taper, £85k–£141.7k	linear / constant 50%	–\$1,408m / –\$932m	§4.1
Reduced rate, £85k–£105k	10% / 15%	–\$398m / –\$199m	§4.1
Bunching readback, 2023–24	$E; b_{\text{LLAT}}$	8,268; 1.411 (target-inherited)	§5

Findings. On the common £85,000 baseline, raising the threshold to £100,000 costs £583m, a taper across the £85,000–£141,667 band costs £1,408m (linear phase-in) or £932m (constant 50% marginal rate), and the 10% and 15% reduced-rate bands cost £398m and £199m. The mechanical anchor reproduces HMRC’s published costing profile to within 30% in every release year, and within 10% in 2025–26 and 2026–27, once band membership is evaluated on data-year turnover; it is an internal consistency check rather than external validation, and the registration behaviour of released firms remains the decisive convention.

The notch creates an exact £21,250 dominated region. Raising the threshold relocates and widens it, and a reduced-rate band divides it between two notches; only the continuous taper removes it. Intensive-margin responses change the reduced-rate costings only modestly under the assumed elasticities, while a threshold rise has no intensive-margin revenue offset in this model.

The bunching exercise supplies a methodological warning, not a behavioural estimate. Its 2023–24 profile is inherited from OBR calibration targets, its degree–window estimates are unstable and mass conservation does not bind. The administrative evidence in [Liu et al. \(2021\)](#) therefore remains the relevant evidence on firm behaviour.

Limitations and future work. The model omits location choice, voluntary-registration choice, compliance costs, sectoral incidence, pass-through, and general-equilibrium effects. Its unregistered stratum has a maintained exponential shape, since the DBT estimates publish only counts and turnover totals by division; threshold cuts, which reach that stratum, inherit the assumption. A fixed cross-section has no registration response or entry margin, which is why the anchor’s first year, where HMRC’s costing carries a registration lag, is the year it fits least well. The nominal-growth factors used for ageing are as-

sumptions recorded in the replication repository. It also cannot capture persistent growth effects extending beyond the cutoff, as documented for a different payroll-tax notch by [Jysmä et al. \(2026\)](#). Those estimates do not transfer directly to UK VAT. Administrative firm-level validation, an explicit registration model, and welfare accounting are the main priorities for future work.

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A Appendix

This appendix collects supporting material in two parts: data construction and calibration robustness (Appendix A.1); and statistical inference for the reduced-form bunching results (Appendix A.2), including the specification-sensitivity grid establishing that band-calibrated synthetic data cannot support bunching inference in either direction.

A.1 Data construction detail

Section 3 describes the two-stage synthetic-population pipeline, parameterised by the registration threshold, and its calibration to HMRC and ONS aggregates. This appendix records the implementation detail deferred from that account.

Each of the approximately 2.94 million firm rows carries a turnover, an input expenditure, a net VAT liability ($v_i = 0.20 (y_i - x_i)$), a calibration weight, a frame indicator, a VAT-scope flag, and a registration flag. The weights are chosen so the weighted frame rows reproduce the ONS enterprise total and employment bands while the weighted registered rows reproduce the HMRC trader counts and liability totals (Table A.1). Because the population is calibrated *to* the HMRC aggregates, agreement between my weighted outputs and those aggregates is a check of *internal consistency*, not external validation; the turnover-band rows in particular hold by construction through the registration propensity. The headline calibration accuracies (98.3% and 98.2% for the two vintages) and the anchor comparison of Section 4 are therefore consistency checks on the pipeline, not out-of-sample tests of firm behaviour.

Table A.1: Calibration accuracy by dimension, $\max\{0, 1 - |\hat{T} - T|/|T|\}$ averaged over the dimension’s targets. HMRC dimensions are scored on VAT-registered rows, ONS dimensions on frame rows. Source: `results/calibration_accuracy.txt`.

Dimension (universe)	2023–24	2024–25
HMRC turnover bands (registered)	100.0%	100.0%
ONS enterprise total (frame)	99.7%	99.8%
ONS employment bands (frame)	99.8%	99.9%
HMRC sector counts (registered)	93.3%	92.6%
HMRC net liability by band (registered)	98.4%	98.9%
Overall (five calibrated dimensions)	98.3%	98.2%
<i>Diagnostic:</i> net liability by sector (registered)	43.1%	44.0%
<i>Diagnostic:</i> net liability below threshold	31.2%	0.0%

The reported calibration score is a bounded summary statistic. For each dimension I compute $\max\{0, 1 - |\hat{T} - T|/|T|\}$, so a dimension missed by more than 100 percent receives zero rather than a negative score. The headline “overall” number is the unweighted mean of the five dimensions used as calibration targets—HMRC turnover bands, ONS population, employment bands, HMRC sector counts, and VAT liability by turnover band. VAT

liability by sector is excluded from that mean and reported separately as an informational diagnostic, because the current generator does not calibrate sector-specific input/output VAT structure.

The replication report also publishes the calibration-weight minimum, median, upper quantiles, maximum, coefficient of variation, and Kish effective sample size. These diagnostics matter because aggregate target agreement can coexist with a small number of influential synthetic records. The report is generated rather than transcribed, and the test suite verifies source-band support, sector-conditional employment assignment, and weighted voluntary registration.

Ledger and Populace migration check. The official target tables are being moved into PolicyEngine Ledger, with Populace providing the shared synthetic-population generator. I therefore keep the paper’s archived processed CSVs as the reproduction source for the reported results, and treat the pinned Populace/Ledger path as an auditable migration check rather than a silent replacement. The snapshot used here is PolicyEngine/ledger pull request 67 at commit `cd98b5c` and PolicyEngine/populace pull request 223 at commit `fa20daf`; both upstream pull requests merged on June 30, 2026. For the 2024–25 vintage, the Ledger-backed targets match the paper’s processed numeric inputs exactly after dropping presentation-only labels, totals, and the HMRC “Unknown” column that the generator does not calibrate: six normalized source tables checked, zero mismatches, and maximum numeric difference zero. (The “2024–25 vintage” here denotes the HMRC/ONS statistical release labelled 2024–25, the same target tables the paper’s 2024–25 population calibrates to.) A 1,000-iteration Populace run from those targets generated 2,946,015 firm rows; its validator reported a 93.8 percent overall score against the paper validator’s then-90.5 percent, a comparison that is not like-for-like (different band sets and sector-target definitions). Two caveats now attach to that pinned snapshot. First, it predates this paper’s net-liability correction (Section 3, footnote): both generators at the pin date shared the mis-scaled $v_i = y_i - x_i$ convention, so the experimental Populace firm generator inherits the same defect, which I have reported upstream; the target-parity result (zero mismatches) is unaffected, since it concerns the input tables, not the generated population. Second, the paper validator’s scores on the current build are 98.3% (2023–24) and 98.2% (2024–25), on the two-universe target layout of Section 3, which the pinned snapshot predates. The comparison table and structured provenance are reproduced by `firm-microsim-populace-ledger` and checked into `results/populace_ledger_comparison.txt` and `results/populace_ledger_provenance.json`.

Counterfactual exclusion window and polynomial degree. The no-VAT counterfactual density of Section 5 is fitted by polynomial regression on the observed density outside a manipulation window around the threshold. The baseline excludes $[T^* - 15k, T^* + 15k]$ and fits a degree-7 polynomial. Widening the window guards against contamination of the counterfactual by the behavioural response but reduces the data available for the fit; raising the degree improves in-sample fit but risks over-fitting the tails. The sensitivity of the excess-mass and bunching-ratio estimates to window width and degree, with associated uncertainty, is reported in Appendix A.2.

A.2 Inference for the bunching estimator

This appendix documents the inference procedure for the reduced-form bunching results of Section 5: the mass-conservation constraint that disciplines the counterfactual density, why no standard errors are reported, sensitivity to the polynomial degree and exclusion window. All numbers are produced by the released estimator run on the re-generated weighted synthetic 2023–24 population; the retained machine-readable artifact is `results/bunching_inference.txt`. The estimator reports the geometry only—the bunching ratio b , its Liu et al. (2021)-normalised counterpart b_{LLAT} , the excess mass E , and the mass-conservation objects (Δ_R, y_R) —and computes no elasticities: the mapping from excess mass to an elasticity is not identified without functional-form assumptions (Blomquist et al., 2021; Bertanha et al., 2023), and on these data there is no behavioural mass to convert in the first place.⁴

Mass-conservation constraint

The canonical bunching estimator of Chetty et al. (2011) and Kleven and Waseem (2013) imposes that the counterfactual density conserves mass: every firm removed from above the threshold reappears as excess mass below it. An unconstrained polynomial fit on bins outside a hand-set window computes the excess mass E (below T^*) and the missing mass Δ_R (above T^*) *independently*, with no guarantee that $E = \Delta_R$ and no endogenous location of the marginal buncher. I impose the constraint in two ways. First, the counterfactual polynomial in $(y - T^*)$ is fit by an iterated procedure in which the density of the bins above the excluded region is rescaled until the fitted counterfactual integrates to the observed total mass over the estimation range [\$20k, \$140k] (the regressor is normalised to $[-1, 1]$ for numerical stability of the high-degree basis). Second, the upper edge of the manipulation region—the marginal buncher y_R —is located endogenously by integrating missing mass upward from T^* until it equals the excess mass below,

$$\int_{T^*}^{y_R} [f^{\text{cf}}(y) - f^{\text{obs}}(y)]_+ dy = \int_{T^*-W}^{T^*} [f^{\text{obs}}(y) - f^{\text{cf}}(y)]_+ dy = E.$$

On the 2023–24 population the baseline gives $E = 8,268$, $\Delta_R = 896$, and $y_R = \$100\text{k}$, the search cap. These mass-conservation outputs are estimator diagnostics, not evidence of a marginal behavioural buncher.

Why no standard errors are reported

Earlier drafts attached bootstrap standard errors to the bunching statistics (resampling firm rows with replacement and re-running the estimator). Those numbers are not reported anywhere in this paper, because the exercise has no estimand: the synthetic file is

⁴Earlier versions of the estimator also printed a CES substitution elasticity and two wedge-normalised turnover elasticities, the latter inheriting an ad hoc $\tau/2$ normalisation from a deleted smoothed-schedule model. These outputs were labelled non-identified but were still reported; they have been removed from the code and the paper.

Table A.2: Headline bunching statistics (point estimates; degree-7 polynomial, $\pm\$15\text{k}$ window). The specification grid characterises sensitivity; standard errors are not reported—see the text.

Vintage	E	Δ_R	b_{LLAT}	y_R (£k)
2023–24 (£85k, OBR fine targets)	8,268	896	1.411	100.0 (censored)
2024–25 (£90k, coarse bands only)	2,475	0	1.445	105.0 (censored)

a deterministic construction from published targets, not a sample from a population, and row-resampling dispersion scales as $1/\sqrt{n}$ in a row count the analyst chooses. A standard error that can be made arbitrarily small by generating more synthetic rows conveys precision that does not exist. The specification grids below instead show how point estimates move with the estimator’s own choices. Generator Monte Carlo error is quantified separately, by rebuilding the full 2023–24 population under alternative generator seeds (`results/seed_sensitivity.txt`): across seeds the half-ranges are ± 198 on E , ± 0.06 on b_{LLAT} , $\pm \$1.7\text{m}$ on the raise costing, $\pm \$3.8\text{m}$ on the taper costing, and $\pm \$0.16\text{bn}$ on the in-scope base—the last larger than in earlier builds because the scope flag is itself a seeded draw within each turnover band. Table A.2 reports the point estimates. Two conventions matter for reading them. Bins are centred on integer £k values (edges at half-integers), so the bin containing the threshold spans £84.5k–£85.5k and its mass is attributed to the above side, attenuating E and Δ_R relative to edge-aligned binning. And E sums the *positive part* of the observed-minus-counterfactual gap in the excluded window below the threshold, so the statistic reads gross rather than net excess.

Sensitivity to polynomial degree and exclusion window

Table entries for a grid of polynomial degree $\in \{5, 6, 7, 8\}$ and symmetric exclusion window $\in \{\$10\text{k}, \dots, \$25\text{k}\}$ are in `results/bunching_inference.txt`; the pattern matters more than the cells. In the 2023–24 grid, E ranges from zero to about 157,600 firms and b_{LLAT} from zero to about 16.5. This specification sensitivity, together with the absence of a location-choice mechanism in the generator, precludes interpreting the baseline as a behavioural estimate.